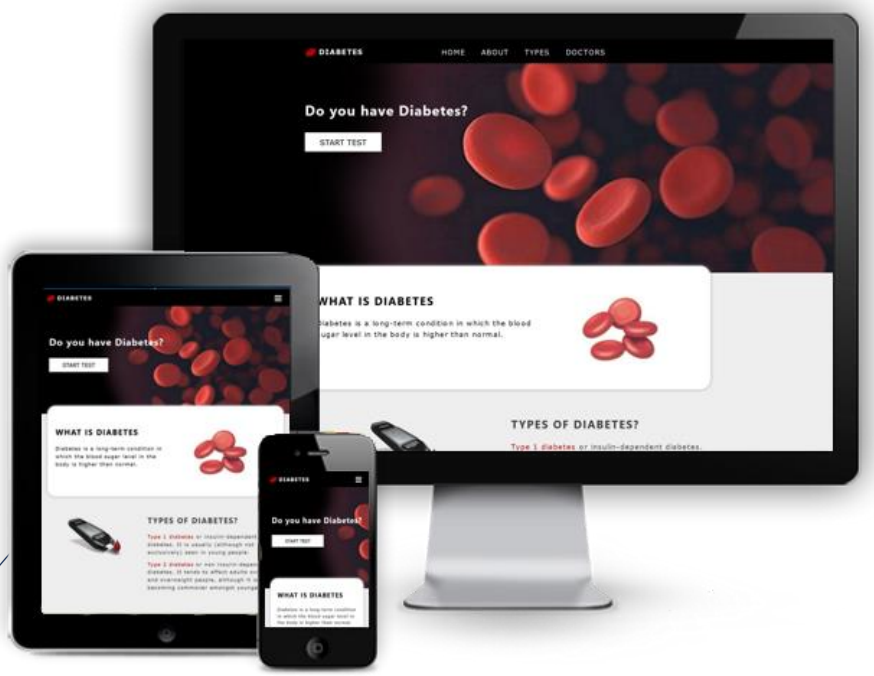
# Development Plan

## Zedland Health Authority

#




Author: Leroy Russell

Website url http://localhost/server/diabetes/index.html

# Content

**Introduction** ..... 2
**System Requirements** ..... 2
Functional requirements ..... 2
Non-functional requirements ..... 2
**Prototyping** ..... 3
Site Map ..... 3
User Interfaces ..... 3
**Implementation** ..... 5
Front-end ..... 5
Back-end ..... 5
**Testing** ..... 5
Compliance ..... 5
Accessibility ..... 6
Browser Compatibility ..... 6
**Enhancements** ..... 6

# Introduction

This report details the basic requirements and design process of the web application developed for Diabetes.

The purpose of the website is to provide user with information about diabetes, calculate user BMI (Body Mass Index) and display result.

# System Requirements

The following functional and non-functional requirements will used for the system.

## Functional requirements

1. Navigation and interactive elements will produce consistent results.
2. System must display each question to user.
3. System must calculate answers provided by user
4. System must display results to user based on the answers given.
5. System must direct user to contact form if result is high.
6. System must display contact form to user.
7. System must validate form field data.
8. System must display appropriate error messages in form.
9. System must submit form if there are no errors.

## Non-functional requirements

1. User interfaces must be aesthetically consistent.
2. System must conform to W3C standards.
3. System must be accessible and useable.
4. System must be fully responsive for desktop and mobile devices.
5. System must provide support for user-agents.

# Prototyping

Prototyping demonstrates the functionality of the system.

## Site Map

The site map shown in figure 1 represents the structure of the web application.

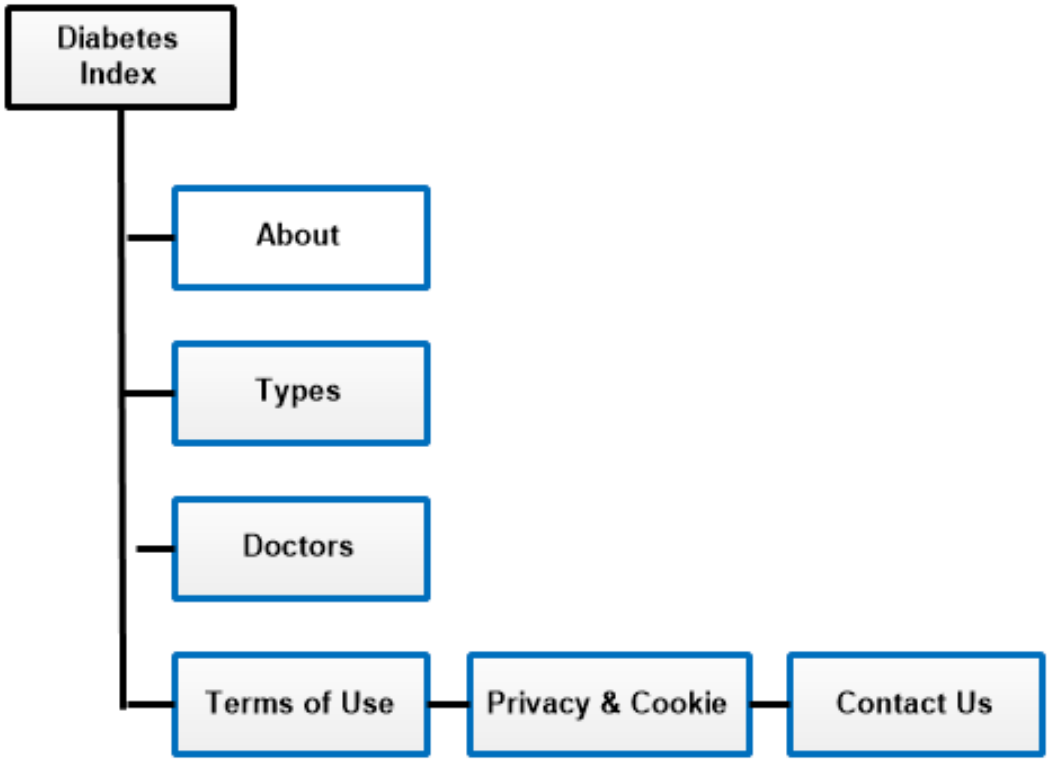



Figure 1. The site map for the web application.

## User Interfaces

The graphical user interface provides the necessary textual and visual information to assist in achieving the required functionality

### Landing/Index Page

The landing page must display information about diabetes and a link for User to take BMI test.

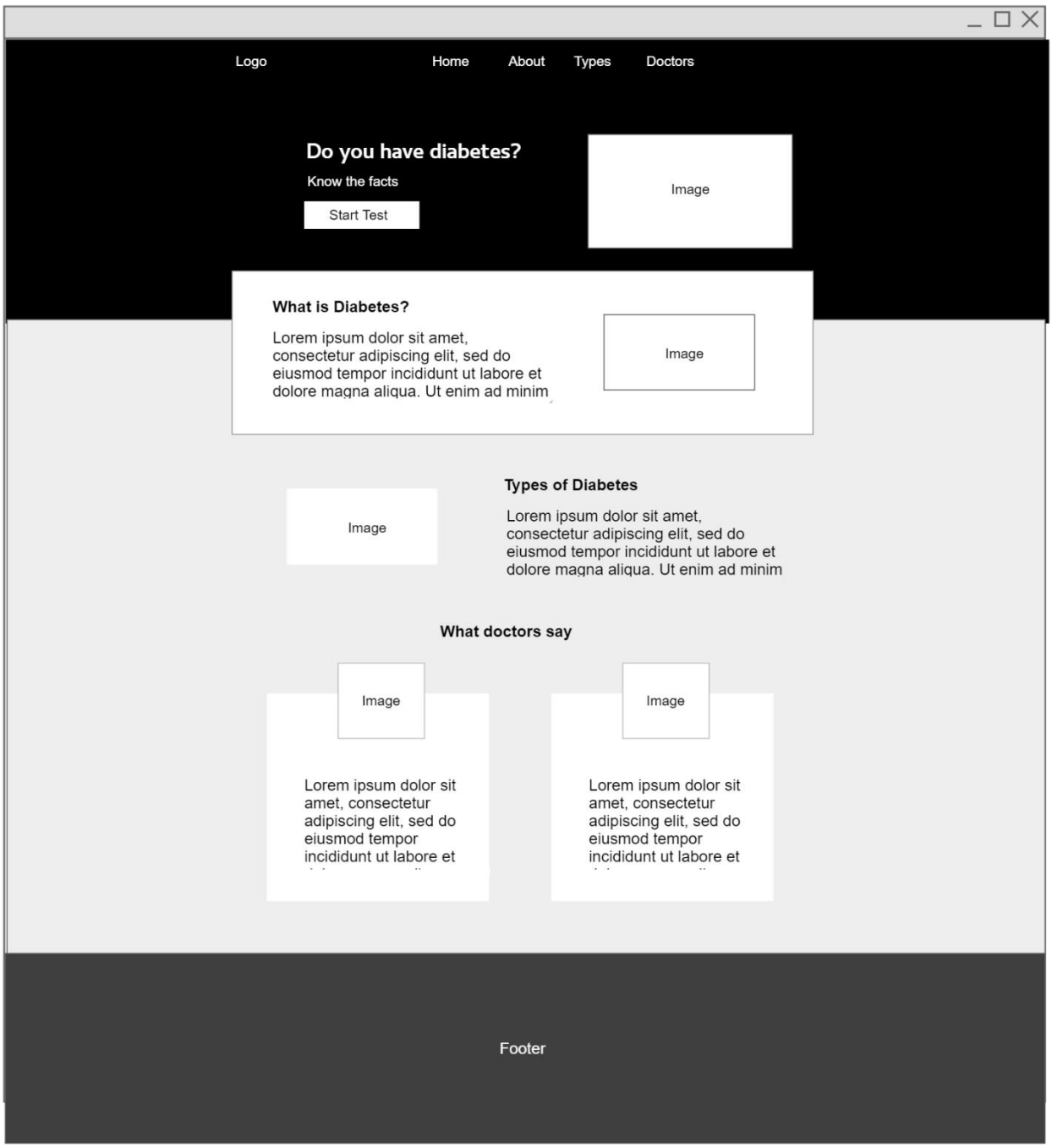



Figure 2. The Landing page for the web application.

# Implementation

The following technologies will be used to build the structure, presentation and functionality of the system.

## Front-end

1. HTML5
2. CSS3
3. JavaScript
4. PHP
5. Krita

| Technology | Usage |
|---|---|
| HTML5 | Structure and layout of the web pages |
| CSS3 | Styling and responsiveness of the web pages |
| JavaScript | Functionality for the website |
| PHP | Server functionality for website |
| Krita | Computer software used for images manipulation and creation |

Table 1. Technologies used for the front-end development.

## Back-end

1. jQuery

| Technology | Usage |
|---|---|
| jQuery | Calculation and validation of user input |

Table 2. Technologies used for the back-end development.

# Testing

The following tests ensure the website meets web standards, browser compatibility and accessibility.

## Compliance

HTML5 and CSS3 have been validated and complies with W3C standards.

### Accessibility

ARIA landmark roles, ARIA properties and skip links have been implemented to enhance website accessibility and usability.

### Browser Compatibility

All web browsers are tested for consistency and responsiveness. The test results are shown in Table 3.

| Web Browser | Version | Result |
|---|---|---|
| Firefox | 79 | Pass |
| Chrome | 84 | Pass |
| Microsoft Edge | 44 | Pass |
| Internet Explorer | 11 | Pass |
| Opera | 69 | Pass |
| Safari | 5.3 | Fail<br>Issue: Alignment and displaying of elements |

Table 3. Test results for web browser compatibility.

## Enhancements

Further improvements for the website are listed in table 4.

| Task | Improvement |
|---|---|
| Implementation | • Eliminate repetition of HTML mark-up<br>• Improve form submission confirmation and URL redirection |
| Browser Compatibility | • Implement HTML5 shiv and other techniques to make website compatible with older browsers |

Table 4. Test results for web browser compatibility.